



**S. B. JAIN INSTITUTE OF TECHNOLOGY,
MANAGEMENT & RESEARCH, NAGPUR.**

Session 2026-27

**LAB MANUAL
Cryptography Lab (N-PECCM501P)**

Year: 3rd

Semester: 5th

Name of Student:

Roll No.:

Semester/Year: Vth semester/3rd Year

Academic Session: 2026-2027

Hardware and Software Requirement

Hardware Requirement

- Processor : Dual Core
- RAM : 1GB
- Hard Disk Drive/SSD : > 80 GB/128 GB

Software Requirement

- Operating System – Windows 2007 and Ubuntu above
- Package used – gcc/g++
- IDE - TURBO C++, DEV C++, Python

Institute Vision:

Emerge as a leading Institute for developing competent and creative Professionals

Institute Mission:

1. Providing Quality Infrastructure and experienced faculty for academic excellence
2. Inculcating skills, knowledge and opportunities for competency and creativity
3. Aligning with Industries for knowledge sharing, research and development

Department Vision:

Become an excellent center for Emerging Technologies in Computer Science to create competent professionals.

Department Mission:

Impart quality education in order to enhance employability and ensure quality teaching-learning process for developing competent professionals with ethics.

Develop advanced technical and professional skills to capture entrepreneurship opportunities in new era of technologies.

Establish the linkages with industries and academia to understand and fulfil the needs of rapidly changing industries and for betterment of society.

Course Outcomes:

On successful completion of this course, students will be able to

- **Understand:** Demonstrate network security protocols such as SSL/TLS, IPsec, and Kerberos for secure authentication and encrypted communication.
- **Apply:** Develop classical and modern cryptographic algorithms, including symmetric and asymmetric encryption techniques, to ensure secure data communication.
- **Analyze:** Assess hash functions, message authentication codes, and digital signature mechanisms to evaluate data integrity and authentication processes.

GENERAL INSTRUCTIONS FOR STUDENTS

DO'S

- Students should enter into the Laboratory with prior permission.
- Students should come in proper uniforms.
- Students should come with Practical note book to the laboratory.
- Students should maintain silence inside the laboratory.
- After completing the laboratory exercise, make sure to shut down the system and arrange chairs properly.

DONT'S

- Students bringing the bags inside the laboratory.
- Students using mobile phones inside the laboratory.
- Students using the computers in an improper way.
- Students scribbling on the desk and mishandling the chairs.
- Students making noise inside the laboratory.

List of Practicals

Aim of Practical		CO Mapped
Pre-Lab	To Understand the fundamentals of Cryptography and explore the virtual lab of Cryptography.	CO1
Practical No. 1	To Create and Apply Classical Cryptographic Techniques: Rail Fence Cipher and Caesar Cipher. The program should provide the user with options to select any of these ciphers to encrypt or decrypt a given message. For each cipher, the user should be able to input the necessary parameters such as keys or keywords, and the program should output the encrypted or decrypted message accordingly.	CO2
Practical No. 2	To apply the Data Encryption Standard (DES) encryption technique to secure a plaintext message. The program should allow the user to input a plaintext message and a 64-bit key for encryption. The program should then output the corresponding cipher text.	CO2
Practical No. 3	Demonstrate the application of RSA Algorithm for encryption and decryption. The program should generate RSA key pairs, encrypt a plaintext message using the public key, and decrypt it using the private key.	CO2
Practical No. 4	Analyze the process of Diffie-Hellman key exchange by implementing it. The program should simulate two parties (X and Y) securely exchanging a shared secret over a public channel.	CO3
Practical No. 5	Analyze the SHA-1 hashing algorithm and evaluate its effectiveness, security, and suitability for different use cases. The program should generate a hash of a given message, also demonstrate if minor change in text/message it will change whole hash.	CO3
Practical No. 6	Examine a basic digital signature system using RSA encryption and a hash function (e.g., SHA) to demonstrate the signing and verification process. The system should be able to: 1. Generate a digital signature for a given message using a private key. 2. Verify the signature using a public key and ensure the message's integrity and authenticity.	CO2,CO3
Practical No. 7	Analyze the MD5 (Message Digest Algorithm 5) hashing algorithm. A college provide a file along with its MD5 hash to ensure safe sharing. Students generate the MD5 hash of the downloaded file and compare it with the given value to check whether the file is unchanged or has been tampered with.	CO3
Practical No. 8	To understand the basic principles of legal and regulatory standards such as GDPR, HIPAA, and PCI-DSS, and explore their applications in identity verification and secure data sharing using simple blockchain-based examples.	CO1
Post-Lab	Open Ended Practical	CO1,CO2, CO3



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Pre-Lab

AIM: To **Understand** the fundamentals of Cryptography and explore the virtual lab of Cryptography.

NAME OF STUDENT	
USN NO.	
SEMESTER/YEAR:	5th Semester/3rd Year
ACADEMIC SESSION:	2026-2027(ODD)
DATE OF PERFORMANCE	
DATE OF SUBMISSION	
TOTAL MARKS (Performance + Viva + Lab Manual Submission)	

AIM: To Understand the fundamentals of Cryptography and explore the virtual lab of Cryptography.

OBJECTIVE/EXPECTED LEARNING OUTCOME:

- To understand the fundamental principles of cryptography.
- To identify various security services such as confidentiality, integrity, authentication, and non-repudiation.
- To explore the features and functionalities of the Cryptography Virtual Lab.
- To familiarize students with cryptographic experiments available in the virtual environment.

THEORY:

Introduction to Cryptography

Cryptography is the science of securing information by transforming readable data (plaintext) into an unreadable format (ciphertext) using encryption algorithms. Only authorized users possessing the correct key can decrypt the ciphertext back into plaintext.

Need for Cryptography

- Secure communication over insecure networks.
- Protection of confidential information.
- Prevention of unauthorized access.
- Verification of data integrity.
- Authentication of users and systems.

Security Services

1. **Confidentiality:** Prevents unauthorized disclosure of information.
2. **Integrity:** Ensures data remains unaltered during transmission.
3. **Authentication:** Verifies the identity of communicating parties.
4. **Non-Repudiation:** Prevents denial of performed actions.

Types of Cryptography

1. Symmetric Key Cryptography

- Same key used for encryption and decryption.

- Examples: DES, AES, Blowfish.

2. Asymmetric Key Cryptography

- Uses public and private key pairs.
- Examples: RSA, ECC.

3. Hash Functions

- Generate fixed-length digest from input data.
- Examples: SHA-256, SHA-512, MD5.

Cryptography Virtual Lab

The virtual lab provides:

- Interactive cryptographic simulations.
- Hands-on experiments with encryption algorithms.
- Demonstration of key generation and digital signatures.
- Analysis of security protocols.
- Practical implementation of cryptographic concepts

Diagram:

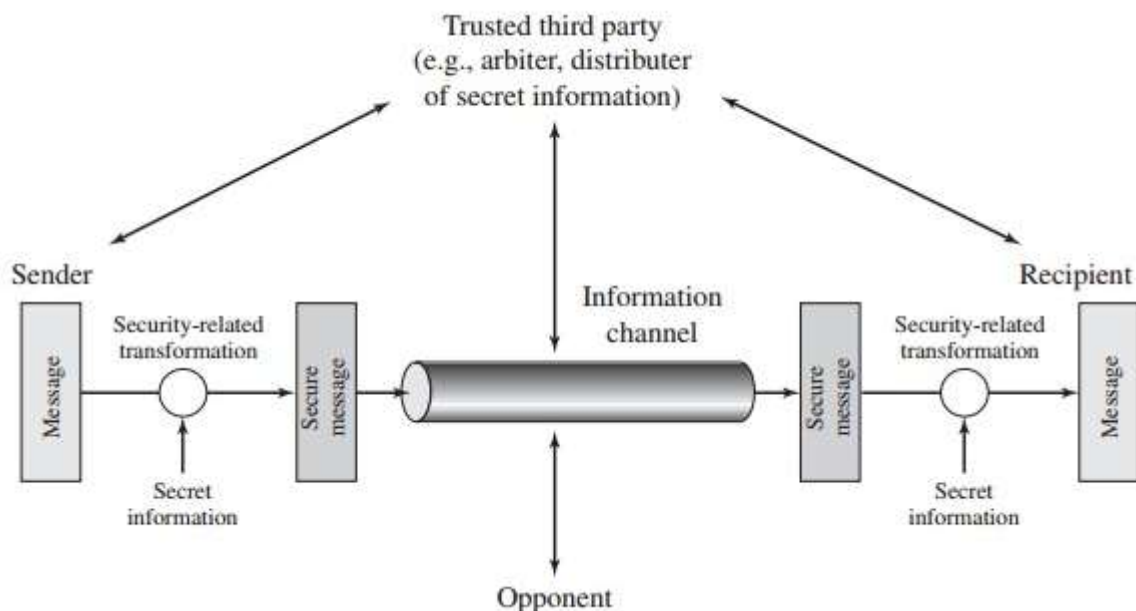


Figure: Network Security Model

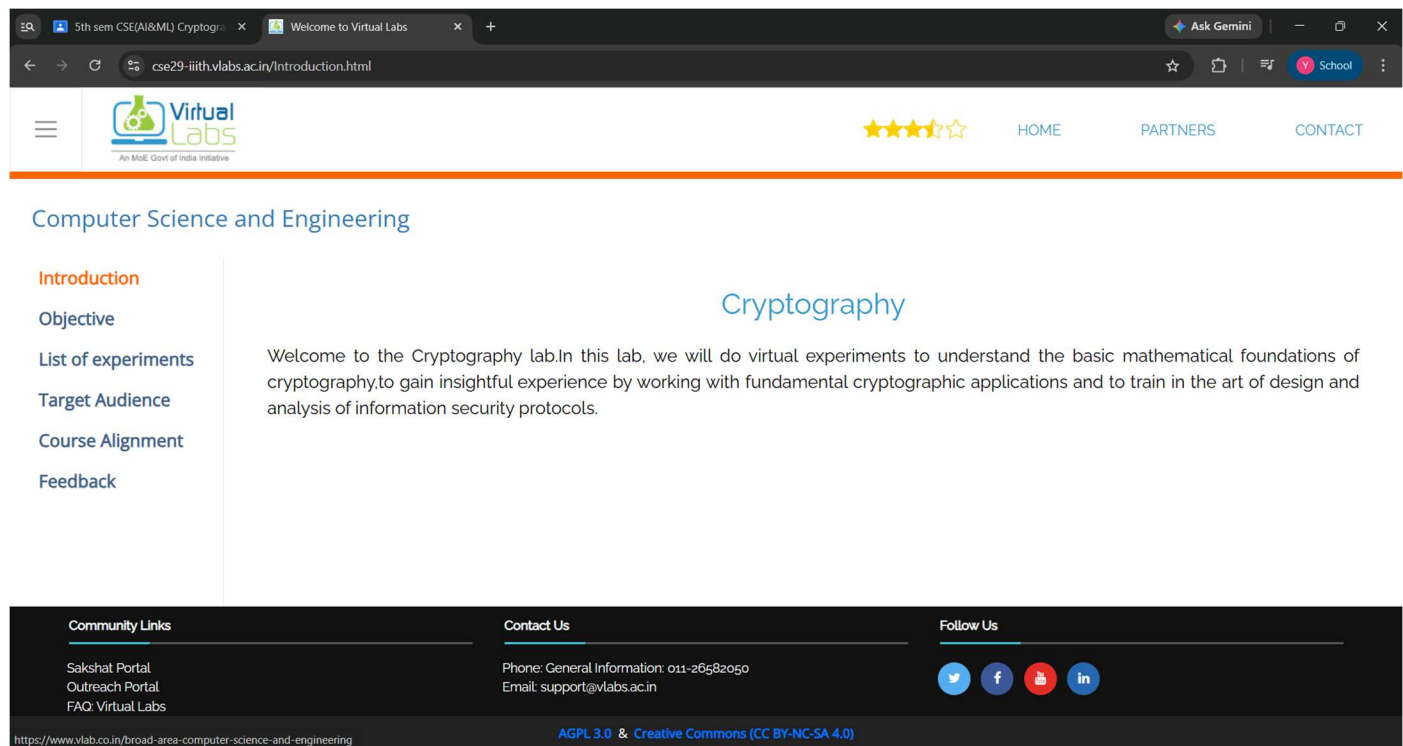
Part A: Study of Cryptography Fundamentals

1. Understand the concept of information security.
2. Study cryptographic terminology.
3. Learn encryption and decryption processes.
4. Analyze various security services.
5. Study different cryptographic algorithms.

Part B: Exploring the Virtual Lab

1. Open the Cryptography Virtual Lab website.
2. Explore the list of available experiments.
3. Study the objectives of each experiment.
4. Observe the simulation environment.
5. Review available learning materials and tutorials.
6. Identify tools used for encryption, decryption, hashing, and digital signatures.

Screenshots:-



Cryptography Lab (N-PECCM501P)

The screenshot shows a web browser with two tabs: '5th sem CSE(AI&ML) Cryptogr...' and 'Welcome to Virtual Labs'. The address bar shows 'cse29-iiith.vlabs.ac.in/Objective.html'. The website header includes the Virtual Labs logo, a star rating, and navigation links: HOME, PARTNERS, and CONTACT. The main content area is titled 'Computer Science and Engineering' and 'Cryptography'. A sidebar on the left lists: Introduction, Objective (highlighted), List of experiments, Target Audience, Course Alignment, and Feedback. The main text states: 'To keep the plaintext secret from eaves- droppers trying to get some information about the plaintext'. The footer contains 'Community Links' (Sakshat Portal, Outreach Portal, FAQ: Virtual Labs), 'Contact Us' (Phone: 011-26582050, Email: support@vlabs.ac.in), and 'Follow Us' (Twitter, Facebook, YouTube, LinkedIn). A license notice 'AGPL 3.0 & Creative Commons (CC BY-NC-SA 4.0)' is at the bottom.

Computer Science and Engineering

Introduction

Objective

List of experiments

Target Audience

Course Alignment

Feedback

Cryptography

To keep the plaintext secret from eaves- droppers trying to get some information about the plaintext

Community Links

Sakshat Portal
Outreach Portal
FAQ: Virtual Labs

Contact Us

Phone: General Information: 011-26582050
Email: support@vlabs.ac.in

Follow Us

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The screenshot shows the same website but with the address bar at 'cse29-iiith.vlabs.ac.in/List%20of%20Experiments.html'. The sidebar now highlights 'List of experiments'. The main content area lists 10 experiments, each with a star rating:

1. Breaking the Shift Cipher ★★★★★
2. Breaking the Mono-alphabetic Substitution Cipher ★★★★★
3. One-Time Pad and Perfect Secrecy ★★★★★
4. Message Authentication Codes ★★★★★
5. Cryptographic Hash Functions and Applications ★★★★★
6. Symmetric Key Encryption Standards (DES) ★★★★★
7. Symmetric Key Encryption Standards (AES) ★★★★★
8. Diffie-Hellman Key Establishment ★★★★★
9. Public-Key Cryptosystems (PKCSv1.5) ★★★★★
10. Digital Signatures ★★★★★

CONCLUSION:

This practical helped in understanding the basic concepts of cryptography, security services, cryptographic algorithms, and the usage of virtual laboratories for performing cryptographic experiments. Students became familiar with the cryptography lab environment and its applications in secure communication systems.

DISCUSSION QUESTIONS:

Q.1) What is cryptography??

Cryptography is the science of protecting information by converting readable data (plaintext) into an unreadable form (ciphertext) using encryption algorithms. It ensures that only authorized users with the correct key can access the original information.

Q.2) What is the difference between encryption and decryption?

Encryption and decryption are two opposite processes used in cryptography to secure data. Encryption is the process of converting readable information (plaintext) into an unreadable form (ciphertext) using an encryption algorithm and a key, so that unauthorized users cannot understand it. Decryption is the reverse process, where the ciphertext is converted back into its original readable form (plaintext) using the correct decryption key. In simple terms, encryption protects the data during storage or transmission, while decryption allows the authorized receiver to access the original information.

Q.3) Define encryption and decryption..?

Encryption:

Encryption is the process of converting readable data (plaintext) into an unreadable format (ciphertext) using an algorithm and a key to protect the information.

Decryption:

Decryption is the process of converting ciphertext back into its original plaintext using the appropriate decryption key.

Q.4) What are the security services provided by cryptography?

Cryptography provides the following security services:

1. **Confidentiality** – Ensures that only authorized users can access the information.
2. **Integrity** – Ensures that the data has not been modified or tampered with.
3. **Authentication** – Verifies the identity of users or communicating parties.
4. **Non-repudiation** – Prevents a sender or receiver from denying that they sent or received a message.

Cryptography Lab (N-PECCM501P)

Q.5) What is the purpose of a virtual laboratory?

A virtual laboratory provides an online environment where students can perform practical experiments without requiring physical laboratory equipment. It helps students understand cryptographic concepts through simulations, practice encryption and decryption techniques, and gain hands-on experience safely and conveniently.

REFERENCE:

- *Cryptography and Network Security, Behrouze A. Forouzan, 1st Edition, 2007, McGraw Hill.*
- <https://cse29-iiith.vlabs.ac.in/List%20of%20experiments.html>
- <https://www.youtube.com/watch?v=JoiLuFNBc4&list=PLBlnK6fEyqRhBsP45jUdcqBivf25hyVkU>

Evaluation:

Viva Grade	
Lab Manual Grade	
Faculty Signature & Date	